



Executive Sales & Operational Dashboard

Business Overview

Café Emily is a café business operating across three locations in New York City: Astoria, Hell's Kitchen, and Lower Manhattan. The business focuses on standard café offerings, including coffee, tea, and baked goods, and primarily serves residents, commuters, and office workers seeking quick, high-quality café options.

The dashboards are designed to assess how the business is performing across these locations and to support decisions related to overall performance and day-to-day operations.

In the short term, the goal is to build a loyal customer base across its existing locations. Over the medium term, the business aims to maintain and continuously improve performance through operational efficiency and consistent service. In the long term, Café Emily plans to expand by opening additional locations across the five boroughs of New York City.

Page One Overview

Who is the primary audience on Page One?

The primary audience for Page One is the Managing Director of Café Emily. The page is designed to provide a clear, high-level view of overall business performance, with a focus on how the business is tracking over time.

What is the single most important message (“Big Idea”) of Page One?

Café Emily is performing well overall, with sales increasing over time. The data indicates that this growth is being driven primarily by higher transaction volumes, while average spend remains relatively stable.

What concrete decision or action should a C-suite executive take?

The positive trends in sales and transactions show that the business is on the right track. Management can continue with the current approach while ensuring capacity is aligned with the areas of strongest demand and use monthly KPIs to spot any early signs of change or risk.

Briefly describe how the order of your visuals creates a logical story from context → insight → action.

The story begins with context, established through the headline KPIs for total sales, total transactions, average spend per transaction, and units sold. These provide a quick snapshot of overall business performance, supported by month filters and previous-month comparisons for quick reference.

This context is extended through the monthly sales trend and forecast, which shows how performance has evolved over time and whether sales are improving, reinforcing the main business narrative.

Insight is then developed by examining how sales are distributed across the business. Sales contribution by location shows that no single store is significantly outperforming the others, suggesting balanced performance across locations. Categories by sales highlights the strongest product drivers as well as underperforming categories. Sales by time of day further refines this insight by identifying peak demand periods.

These insights naturally lead to action, as understanding where and when sales are strongest supports decisions around operational focus. Identifying peak morning and midday periods provides guidance for aligning capacity and resources to match demand and maximise performance during high-traffic windows.

Page Two Overview

Who is the intended operational audience for Page Two?

The intended audience for Page Two is the Operations Manager who oversees all café locations. Given the size of the business, a single manager is responsible for day-to-day operations across the cafés, with no need for separate regional management due to the proximity of the café locations.

For each drill-through field, describe how the detail page continues the story from the summary page.

Page Two continues the story from the summary page by using drill-through to move from overall performance to operational detail. While Page One shows that sales growth is driven by increasing transaction volumes, this page explains where, when, and how that demand occurs.

Store Location drill-through

The store location drill-through builds on the summary view by moving from location-level performance to operational behaviour. While Page One shows that sales are relatively balanced across café locations, drilling through to a specific store reveals when activity is highest and how demand is distributed across days and hours. This shows how each location supports overall performance and reveals demand differences hidden at the summary level.

Product Category drill-through

The product category drill-through builds on the summary by breaking performance down across each product category offered by Café Emily. When a category is selected, the detail page shows a full table of all menu items within that category and how each contributes to sales and transactions. The price and volume context helps explain how revenue is being generated within the category and links back to the Transaction Volume and Sales chart, where sales generally follow transaction volume rather than pricing changes.

Time of Day drill-through

The time-of-day drill-through builds on the summary by focusing on a specific period and filtering the page to that window. It shows the peak business hours and how transaction activity is concentrated during that period, with the heat map providing weekly context and the table showing which menu items drive sales and transactions at that time.

Finally, the operational page is designed to stand on its own. When viewed independently, the location and month filters still provide a clear and detailed operational view of how the café is performing.

What new question does the detailed visual answer that could not be answered from Page One alone?

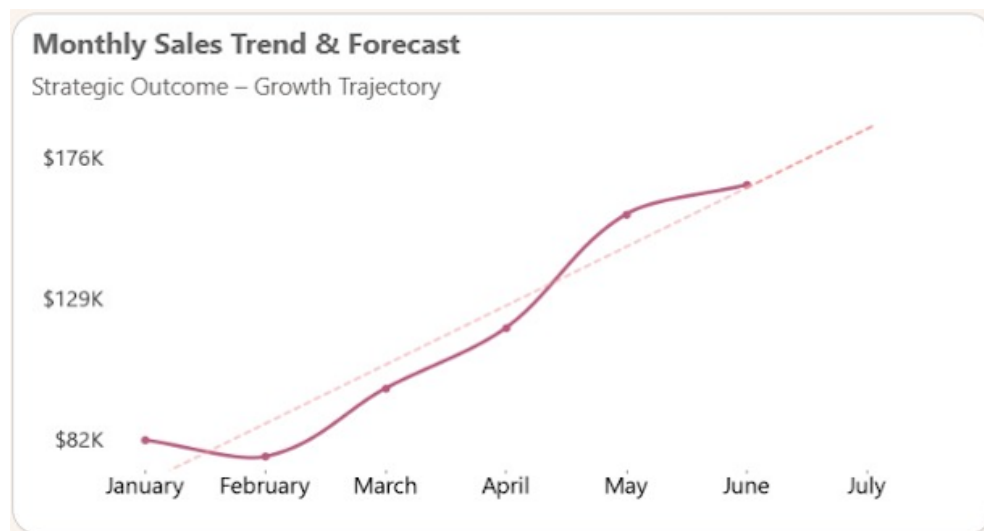
Page Two answers how sales growth happens in practice by addressing questions that cannot be answered from the executive page alone. While the summary KPIs such as average spend and units sold show overall performance, the detailed visuals break this down to menu item and time-of-day level. This allows us to see individual item performance, identify which menu items dominate during different hours of the day, and understand whether overall revenue is being driven by transaction volume or by pricing.

Design Decisions

For one key visual on Page One and one on Page Two, explain why that chart type is the best way to tell this part of the story.

Page One

Figure 1: Monthly Sales Trend & Forecast Line Chart



The line chart represents the trend in sales for the first six months of 2023. It intuitively communicates month-by-month changes in sales, allowing trends to be understood immediately. Peaks represent periods of growth, while troughs indicate declines in performance. The line chart also helps executives assess expected future sales performance, as one of its strengths is highlighting underlying trends in the data. In addition, it supports the identification of business risk, such as low or declining sales months, and enables strategic forecasting by allowing historical patterns to be analysed and used to inform future decisions and opportunities. In summary, the Monthly Sales Trend and Forecast line chart emphasises movement and direction

over time. It highlights growth opportunities and risk periods, while also supporting predictive analysis and strategic planning.

Alternative Graph – Bar Chart

A bar chart is well suited for comparing discrete data instead of continuous data illustrated by the line chart. A bar chart does not do a good job of communicating momentum or trend continuity.

A line chart is better suited for trend analysis as a bar chart would fragment the growth narrative of the data story being told.

Page Two

The combined line and column chart highlights the relationship between sales and transactions volume with respect to time, making it clear that sales are volume- driven. A line chart helps to highlight differences and correlations within data over time.

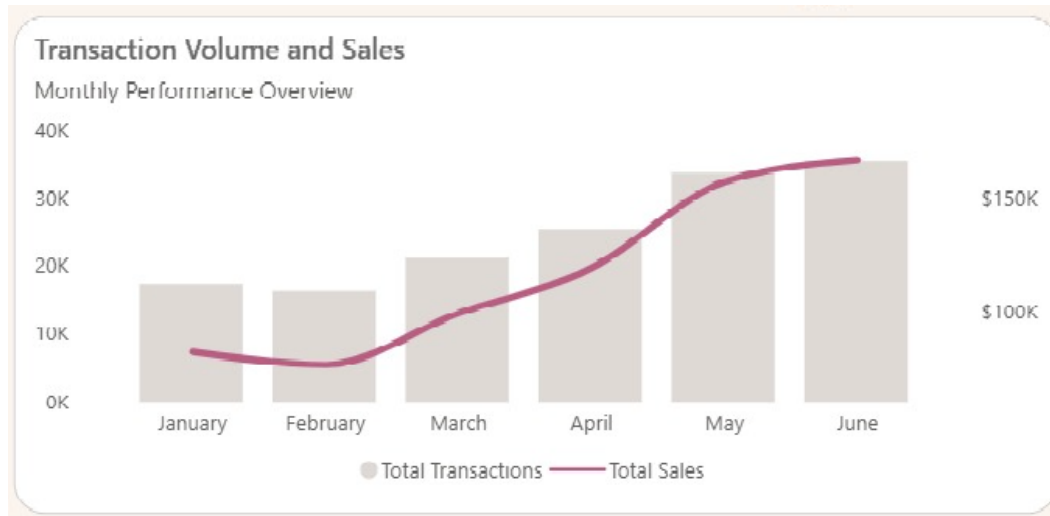


Figure 2: Transaction Volume and Sales Combined Line and Column Chart

In summary, the Transaction Volume and Sales combined line and column chart shows the relationship between sales and transactions. It maintains the time context for both variables, allowing their movement to be compared over the same period.

Alternative Graph – Scatter Plot

Scatter plots are best used to show correlation between variables by visually illustrating how two or more variables are related. While they are effective at revealing positive or negative relationships, they lack a time axis and therefore cannot show monthly performance trends for sales and transaction volume, making them unsuitable for operational monitoring.

Describe two specific clutter-reduction decisions you made and how they improve understanding.

1. Removal of gridlines and excess labels

Gridlines and unnecessary labels were removed to reduce visual noise and keep attention on the data itself. This makes the charts easier to read and allows key patterns to be understood immediately, which is especially important for executives who need to interpret information quickly.

2. Simple colour palette

Likewise, using simple colours that are not too bold or bright helps viewers focus their attention on the main idea each visual is meant to communicate. The dashboard maintains a professional and clean look when simple colours are used, making the information easier to interpret.

Describe two specific ways you direct attention.

Colour

A warm, café-inspired colour palette was derived from the Café Emily logo and used deliberately to direct attention rather than for decoration. Distinct colours are applied in the Monthly Sales Trend and Forecast line chart on Page One, with Burgundy/Plum (#B95E82) used for the historical sales trend and Soft Rose (#F39F9F) for the forecast, helping viewers distinguish between actual performance and projected movement. Throughout the dashboard, colour is used purposefully, with no insight relying on colour alone to convey meaning. Red and green indicators are restricted to KPI visuals, such as month-over-month arrows, while charts primarily use neutral or unsaturated tones, reserving brand colours for emphasis and focal points.

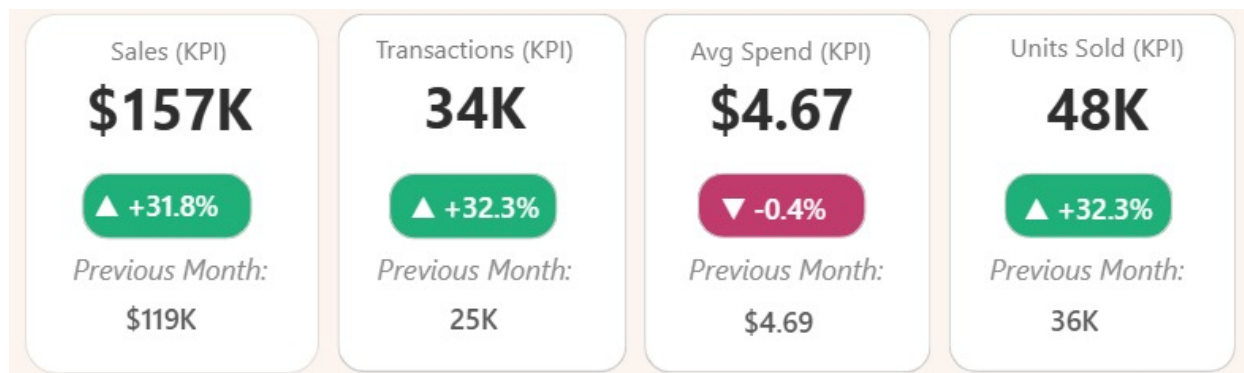
Figure 3: Colour Palette



Font

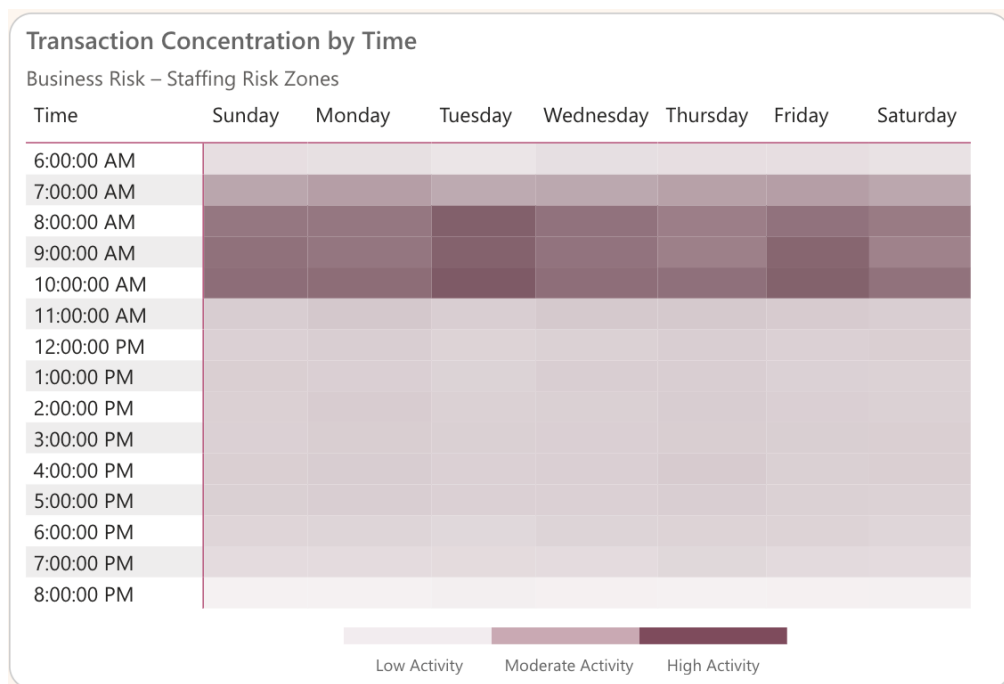
The dashboards were designed using the Segoe UI font family throughout. Visual hierarchy is created through text size, with bold used for titular elements, semi-bold for chart titles, and regular weight for subtitles. The executive KPIs were formatted accordingly to highlight the most critical figures for strategic decision-making. Using bold and italicised text reduces distraction from information that is not critical for decision-making.

Figure 4: Executive Sales Dashboard (2023) – KPIs



Choose one visual and explain: How does it contribute to the overall story about Strategic Outcomes, KPIs, or Business Risk, rather than just “showing data”?

Figure 5 : Transaction Concentration by Time (Heatmap)



The heat map contributes to the overall story by doing more than just showing transaction counts. It highlights when activity is concentrated across days and hours, making periods of high and low demand easy to identify. This helps the Operations Manager see where operational risk exists, such as peak times that may require additional staffing, as well as quieter periods where resources may be underutilised. In doing so, the heat map supports practical operational decisions around staffing and daily store management, rather than simply displaying activity levels.

Data Overview & Limitations

The analysis is based on a publicly available café transaction dataset sourced from Kaggle (<https://www.kaggle.com/datasets/agungpambudi/trends-product-coffee-shop-sales-revenue-dataset>). The dataset contains transaction-level records, including date and time of purchase, menu items, prices, quantities, and store locations.

As the data is sample data and not taken from a real business, the Café Emily brand and business context were created around the dataset to support a coherent and realistic data story. Analytical and design decisions were made to reflect how an actual café operation might be assessed, focusing on sales performance, transaction volume, product mix, and time-based demand. This allows the dashboards to demonstrate how data can be used to evaluate performance, identify operational risks, and support decision-making.

While the dataset is suitable for illustrating analytical and visualisation techniques, it does not capture all real-world factors such as external market conditions or customer demographics, and results should be interpreted within this context.